

Approaches to learning

Approaches to learning

The development of skills such as thinking skills and communication skills is frequently identified as a crucial element in preparing students effectively for life beyond school. A 2007 survey of 400 hiring executives of major US corporations identified their top four requirements of new recruits as being oral and written communication skills, critical-thinking and problem-solving skills, professionalism and work ethic, and teamwork and collaboration skills (Trilling and Fadel 2009). Similar skills lists have been developed by the International Society for Technology in Education (ISTE) and the Organisation for Economic Co-operation and Development (OECD), and are also the subject of numerous books such as *The Global Achievement Gap* (Wagner 2010). Yet many students struggle with some of the basic skills of effective learning. For example, good note-making has been positively correlated with academic achievement, yet research suggests that many university students have difficulty even with the fundamental skill of making notes from lectures or texts (Kiewra 1985, O'Donnell and Dansereau 1993).

Developing students' ATL skills is about more than simply developing their cognitive skills. It is also about developing affective and metacognitive skills, and about encouraging students to view learning as something that they "do for themselves in a proactive way, rather than as a covert event that happens to them in reaction to teaching" (Zimmerman 2000: 65). By developing ATL skills and the attributes of the learner profile, DP students can become "self-regulated learners" (Kaplan 1998). Self-regulated learners have learned how to set learning goals, ask good questions, self-interrogate as they learn, generate motivation and perseverance, try out different learning processes, self-monitor the effectiveness of their learning, reflect on achievement, and make changes to their learning processes where necessary (Zimmerman and Schunk 1989, de Bruin et al. 2011, Wolters 2011).

The term "skill" is therefore used in a broad sense in the DP to encompass cognitive, metacognitive and affective skills. Cognitive skills include all the information-processing and thinking skills, often called "study skills" in a school environment. Affective skills are the skills of behaviour and emotional management underpinning attitudinal factors such as resilience, perseverance and self-motivation, which often have a large role to play in educational achievement. Metacognitive skills are the skills that students can use to monitor the effectiveness of their learning skills and processes, to better understand and evaluate their learning. Although these skills may be in use when manifesting a

certain natural ability or talent, they are different to both of these because proficiency in any skill can be increased through the deliberate use of techniques and strategies, feedback and challenge. Skills are therefore highly teachable.

In the DP, as well as in the Primary Years Programme (PYP) and Middle Years Programme (MYP), these cognitive, metacognitive and affective skills are grouped into the same five ATL categories.



Although these skills areas are presented as distinct categories, there are obviously close links and areas of overlap between them, and it is intended that these categories should be seen as interrelated. It is also the intention that these ATL skills should be seen as linking closely with the attitudes and dispositions identified in the IB learner profile. The learner profile is the [IB mission statement](#) translated into a set of learning outcomes for the 21st century. It is an easily communicated set of ideals that can inspire, motivate and focus the work of schools and teachers, uniting them in a common purpose.

The next five sub-sections of this document will explore each of the five ATL skills categories in turn. They will identify some of the specific skills that make up these categories, discuss what these skills look like in students, and discuss strategies for their development.

Related resources

[Video: Self-management: self-directed learning \(Vienna\)](#)

Name of school: Vienna International School, Austria

DP subject: Language B German lesson

Related resources

- [Language B German lesson](#)

[Video: Communication skills \(Bangkok\)](#)

Name of school: NIST, Bangkok, Thailand

DP subject: Language A literature lesson

[Video: Communication Skills \(Sevilla\)](#)

Name of school: Colegio de San Francisco de Paula, Sevilla, Spain

DP subject: Mathematics lesson

Thinking skills

“Concern with developing students’ thinking, far from being a fad, is one of the most persistent and ambitious aspirations of education.”

— (Swartz and Perkins, 1989: 5)

Developing thinking skills is a key feature of the constructivist approach that so heavily influences all IB programmes. In this approach, the teacher is seen as a facilitator who “guides the student, stimulating and provoking the student’s critical thinking, analysis and synthesis throughout the learning process” (Briner 1999: 1). Being “thinkers” is explicitly identified as one of the IB learner profile attributes, and is defined in terms of exercising initiative in applying thinking skills critically and creatively to recognize and approach complex problems, and make reasoned, ethical decisions.

The DP prides itself on providing opportunities for students to develop thinking skills and also an awareness of themselves as thinkers and learners; something that is most explicitly apparent in the

important place given within the DP to the theory of knowledge (TOK) course. TOK plays a special role in the DP, providing an opportunity for students to reflect on how we know what we claim to know. It is intended that the study of TOK will support, and also be supported by, the academic disciplines, and all DP subject guides provide teachers with suggestions for how they can make links to TOK in their subject.

Teaching idea

As part of their TOK course, DP students compare and contrast the methods used to gain knowledge in different areas (such as the natural sciences, history or the arts). Initiate a discussion with your students about the methods used in your DP subject. What are the methods used in your subject, and what is it about these methods that generates knowledge? What are the assumptions underlying these methods? What are the limitations/constraints on these methods?

The term *thinking skills* refers to a cluster of a large number of related skills, and in the DP particular focus is placed on skills such as metacognition, reflection and critical thinking. Many categorizations of thinking skills also emphasize related behaviours such as “curiosity, flexibility, posing problems, decision making, being reasonable, creativity, risk taking, and other behaviours that support critical and creative thought” (Costa and Kallick 2009). In this way, while a crucial element of developing students’ thinking is helping them to build up a toolkit of skills, it is equally important to cultivate dispositions such as open-mindedness and curiosity.

Whether metacognition is a skill, a collection of skills or simply a state of mind is debatable, but in essence metacognition “refers to higher-order thinking which involves active control over the cognitive processes engaged in learning” (Hattie 2009). In many ways, improving metacognition is foundational for improving all other skills. Improving students’ metacognitive awareness will help them to become more aware of the ways in which they process information, find patterns, build conceptual understandings, and remember key facts and ideas. Once they are aware that they are using techniques and strategies to perform even the most basic learning tasks, students can be encouraged to consider if there are more effective or efficient ways to achieve those same outcomes. They can then be encouraged to try new ways of learning and to evaluate the results.

Reflection

Like metacognition, reflection is a thinking skill that plays a particularly essential role in learning. The value of structured reflection activities in education is widely recognized, and elements are commonly incorporated into learning programmes in various forms. These activities may be written (for example, reflective essays, reading logs, case studies), oral (for example, presentations, video diaries, discussion) or artistic/creative (for example, photos or other images capturing scenes that can then be used to promote discussion). They may also encompass elements of more than one form (for example, student portfolios). Online tools such as blogs (reflective journals), wikis (collaborative web pages), digital stories and podcasts, interactive micro-blogging and social networking also offer great opportunities for reflective activities.

It is crucial that reflective activities are meaningful for students, and do not become mechanistic and formulaic. Effective reflective activities should challenge students to think more deeply, and students need to be explicitly aware of the role of the activities. This does not necessarily mean introducing students to specific theories/models of reflection, although this can be helpful, but it does mean ensuring that sufficient time is allocated to appropriately designed reflective activities at all stages of learning—before, during and after. Another helpful approach can be to distinguish between different types of reflection. For example, the three elements of the DP core all have a slightly different emphasis in relation to reflection: in TOK, the prime emphasis is on cognitive reflection; in the extended essay, the emphasis is on reflection on process; and in creativity, activity, service (CAS), there is a more affective emphasis to the reflection.

Higher order thinking

Discussions and categorizations of thinking skills often use the phrase *higher-order thinking skills*. This distinction between higher- and lower-order thinking skills is rooted in Bloom's taxonomy of thinking skills (1956). Bloom's taxonomy distinguishes between the lower-order skills of knowledge acquisition, comprehension and application, and the higher-order skills of analysis, synthesis and evaluation. Bloom's taxonomy was later revised and updated by Anderson and Krathwohl (2001) into a slightly less strict hierarchy that allows for more overlap between the categories.

Category	Associated thinking skills
1. Remember	Recognizing, recalling
2. Understand	Interpreting, exemplifying, classifying, summarizing, inferring, comparing, explaining
3. Apply	Executing, implementing
4. Analyse	Differentiating, organizing, attributing
5. Evaluate	Checking, critiquing
6. Create	Generating, planning, producing

(Anderson and Krathwohl, 2001)

DP subjects place a premium on the development of higher-order thinking skills rather than on simple memorization of content. This is clearly apparent in the aims of DP courses; for example, one of the aims of all group 3 courses is to “develop in the student the capacity to identify, to analyse critically and to evaluate theories, concepts and arguments about the nature and activities of the individual and society”. The higher-order thinking skills DP subjects foster can be seen in the command terms used in DP examination papers, where questions ask students to “analyse”, “evaluate”, “compare”, and so on. Most DP subject guides provide a list of command terms, and it is important that teachers familiarize students with these terms so that students have a clear understanding of what they are expected to demonstrate in their responses.

Teaching idea

When planning DP lessons, consider what higher-order questions you will ask your students to encourage higher-order thinking. Also be sure to plan time for students to think about their answers to questions, rather than engaging in rapid “ping-pong” questions and answers that do not give students time to think deeply about their responses.

Some of the most influential recent work on thinking comes from the “Project Zero” team at the Harvard Graduate School of Education. Their “Cultures of Thinking” project focuses on the importance of creating an atmosphere in the classroom where the “group’s collective as well as individual thinking is valued, visible, and actively promoted as part of the regular, day-to-day experience of all group members” (Ritchhart et al. 2011). This approach emphasizes the importance of embedding thinking into the culture and everyday life of the school, rather than it being seen as an add-on. One practical teaching strategy developed by this project to help achieve this aim is that of “visible thinking” routines (Ritchhart et al. 2011), for example:

Connect, extend, challenge (Routine to help students make connections between old knowledge and new knowledge.)	Generate–sort–connect–elaborate (Concept-mapping routine.)	I used to think ... now I think ... (Routine to help effective reflection on how and why their thinking has changed.)
Headlines (Routine where students write a newspaper-style headline to capture the essence of an idea, event, topic, etc.)	Claim, support, question (Routine for exploring claims. Students make a claim, identify support for that claim, then ask a question related to that claim.)	Think, pair, share (Routine where students think individually, then share/compare their thinking with a partner.)

Related resources

[Interview: Robert Swartz](#)

[Video: Thinking skills: critical thinking \(New York\)](#)

Name of school: UNIS, New York City, USA

DP subject: Biology lesson

Related resources

- [Queen Victoria's Pedigree Chart: Deducing genotypes and phenotypes](#)
- [Queen Victoria's Pedigree Chart: Deducing genotypes and phenotypes \(using 1 to 1 computers\)](#)

[Video: Thinking skills: visible thinking \(Prague\)](#)

Name of school: International School of Prague, Czech Republic

DP subject: Language B Mandarin ab initio lesson

Related resources

- [Cooking video homework](#)
- [Mandarin thank you letter](#)
- [Restaurant assessment task](#)

Communication skills

"Most people do not listen with the intent to understand; they listen with the intent to reply."

— (Stephen Covey, 2004: 239)

Surveys asking hiring executives which skills they are looking for in potential future employees frequently report oral and written communication skills as being top of the list (see, for example, the survey in Wagner 2010). Students in the final years of their school-based education need to be very aware of this link, as the requirement for strong communication skills is often implied or explicitly identified in job descriptions (Krapels and Davis 2003).

Within DP schools, good communication skills are needed not only for success in every subject discipline but also to help form and maintain good interpersonal relationships both with other students and with adults, be they teachers, administrators or parents (Gallagher 1991). Good communication skills have also been significantly correlated with improved social self-efficacy (Erozkan 2013), self-esteem and self-confidence (Riggio, Throckmorton and DePaola 1990).

The term *communication skills* is very broad and consists of a cluster of different skills and forms of communication. The University of Melbourne, in association with Cisco, Intel and Microsoft, produced an influential white paper as the foundation document for their “Assessment and Teaching of 21st Century Skills” (ATC21S) project, in which they state that the most essential communication skills for adolescents are as follows (Binkley et al. 2010):

- The ability to communicate, in written or oral form, and understand, or make others understand, various messages in a variety of situations and for different purposes.
- The ability to listen to, and understand, various spoken messages in a variety of communicative situations, and to speak concisely and clearly.
- The ability to read and understand different texts, adopting strategies appropriate to various reading purposes (reading for information, for study or for pleasure) and to various text types.
- The ability to write different types of texts for various purposes. To monitor the writing process (from drafting to proofreading).
- The ability to formulate one’s arguments, in speaking or writing, in a convincing manner and take full account of other viewpoints, whether expressed in written or oral form.
- The skills needed to use aids (such as notes, schemes, maps) to produce, present or understand complex texts in written or oral form (speeches, conversations, instructions, interviews, debates).

All of these communication skills play a crucial role in the DP. For example, within the three elements of the DP core, students need to communicate in a number of different ways: in TOK, students undertake both an essay and an oral presentation, and need to be able to formulate arguments clearly and coherently in both formats; in the extended essay, the skill referred to above of monitoring the writing process from drafting to proofreading is particularly vital; and in CAS, students often, for example, undertake service activities that require effective interaction and communication with members of the local community.

Teaching idea

When supporting students in planning their CAS activities, encourage them to carefully plan and reflect on the ways in which they will communicate with the people and communities with whom they will be working. Encourage them to think about what the potential challenges might be, as well as the positive opportunities for using their communication skills through activities such as advocacy.

Strategies to develop communication skills

There are many simple strategies that teachers can use to help students develop their communication skills. For example, Mendler (2013) identifies eight simple tips for encouraging good communication in the classroom:

- Model a good conversation, particularly with students who struggle conversationally.
- Encourage physical cues.
- Challenge put-downs or hurtful comments.
- Ask open-ended questions.
- Put thinking ahead of knowing.
- Have informal conversations.
- Make eye contact.
- Encourage turn-taking.

(Mendler, 2013)

The link between communication and social skills

Communication is a two-way process that involves an exchange of ideas and information. This means that there is a particularly strong link between communication skills and social skills, which are the subject of the next sub-section of this document. For example, for many teenagers, the internet means connection with others—both communication and social interaction—whether by email, Facebook, Twitter, chat, blogs, gaming, and so on. This is today's participatory social culture where creativity, communication and collaboration, blended with media skills, are the currency of value. These skills

are sometimes called “media literacy”. In the US, studies show that 57% of online teenagers (around 12 million people) create content for the internet. They create blogs and web pages for school, friends or organizations, they share original artwork, stories, photos or videos online, or they sample and remix other online content into new original creations. Lenhardt and Madden (2005) have identified that:

- 33% of online teens share their own creations online, such as artwork, photos, stories, or videos
- 32% say that they have created or worked on web pages or blogs for others, including those for groups they belong to, friends or school assignments
- 22% report keeping their own personal web page
- 19% have created their own online journal or blog, and 38% read them.

Online activities such as these can present exciting opportunities for the development of communication skills.

Related resources

[Video: Communication skills \(Washington DC\)](#)

Name of school: British School of Washington, USA

DP subject: Language A literature lesson

Related resources

- [Communication skills at the British School of Washington](#)

[Video: Social and Communication Skills \(Singapore\)](#)

Name of school: Tanglin Trust School, Singapore

DP subject: Language A literature lesson

[Case study: The Jane Schaffer Paragraph—developing organized writing across the curriculum](#)

Social skills

To function effectively in the school environment, students need to be adept at both peer-related and teacher-related social communication and behaviour. These skills are closely linked to communication skills and also to attributes of the IB learner profile, such as being caring (for example, through students being aware of the impact of their behaviour on others).

Learning itself can also be seen to have a strongly social element. This idea is particularly associated with Vygotsky, who argued that “human learning presupposes a specific social nature and a process by which children grow into the intellectual life of those around them” (Vygotsky 1978: 88). In this approach, learning is a fundamentally active social process, and collaboration is a crucial way of constructing understanding and making meaning. (For further discussion of this pedagogical approach please see the [“Teaching focused on effective teamwork and collaboration”](#) section of this document.)

Specific training in social skills can have a number of other benefits, particularly for students with some form of pre-existing behavioural difficulty. Social skill deficiency in childhood has been found to be the single best predictor of significant problems in adulthood (Strain and Odom 1986) and it is more frequently found in teenagers with learning and behavioural problems than in the general student population (Elksnin and Elksnin 1998). In a 1991 study, 91% of a group of 99 social phobics reported adverse effects on their academic performance as a result, usually through lack of class participation, fear of assignments requiring oral presentations and group work (Turner et al. 1991).

Teaching idea

Promote appreciation of individual differences.

The IB mission statement encourages students to “understand that other people, with their differences, can also be right.” Discuss with students why the IB might have included this in its mission statement, and discuss scenarios they have encountered/are likely to encounter where this skill is important.

The term *social skills* encompasses a broad cluster of skills; Walker (1983), for example, defines social skills as “a set of competencies that a) allow an individual to initiate and maintain positive social relationships, b) contribute to peer acceptance and to a satisfactory school adjustment, and c) allow an individual to cope effectively with the larger social environment” (Walker 1983: 27). However, a particularly crucial skill, within the category of social skills, is collaboration.

Collaboration

Collaborative learning has been found to improve understanding of facts (Fall et al. 1997), increase student motivation and engagement (Cumming 2010), improve academic performance for lower-ability students (Saner et al. 1994), and to produce higher satisfaction in learning (Klein 1992).

Another of the key advantages of collaborative activities is that they can be a catalyst to higher-order thinking. “Proponents of collaborative learning claim that the active exchange of ideas within small groups not only increases interest among the participants but also promotes critical thinking” (Gokhale 1995). Discussion and argument are key aspects of both critical thinking and many collaborative tasks. Generating creative ideas and problem-solving are also two examples of tasks where students are often stimulated to perform better when they engage in collaborative reflection. Johnson and Johnson (1983) “found that when students work cooperatively in groups, increased reasoning strategies and greater critical thinking competencies result than in competitive or individualistic settings” (in Costa and Lowery 1989: 17).

There is evidence that cooperative teams achieve at higher levels of thought and retain information longer than students who work quietly as individuals. In some cases, “working with others promotes academic engagement through the added responsibility of group performance, which causes individuals to persist at difficult tasks longer than they normally would (Lai 2011: 22). Collaborative activities give students an opportunity to engage in discussion and to be exposed to alternative points of view, which means that “cooperative learning and critical reflection are natural allies” (Kagan 2003).

Teachers should provide explicit opportunities in the classroom for students to practise and develop their social and collaborative skills. Group formative assessment tasks can provide an excellent opportunity and incentive for students to improve their collaboration skills.

An example of a method that can be used to help plan effective collaborative tasks is the spider web discussion method created by Alexis Wiggins (2011). The name is an acronym, describing the specific aspects of the group task and its process.

synergetic—a collaborative, group effort with a single group grade

process—a process that must be practised and honed

independent —students work independently; teacher observes and gives feedback

developed—a developed, sustained discussion that aims to “get somewhere”

exploration—an exploration of ideas, texts or questions through discussion with a ...

rubric—a clear, specific rubric against which the students can self-assess.

(Wiggins, 2011)

The word “web” then describes two aspects of the method. First, it describes the physical map of the discussion, which looks like a web. Second, it is a metaphor for the process—like a web, all participants must pull their own weight equally, or the web cannot be strong. Through specific processes such as modelling, coding, group grading and feedback sessions, activities such as these “train students to work together collaboratively in their problem-solving and to self-assess that process. The result is deep, high-level inquiry led and assessed by the students themselves, and the creation of authentic collaborators, communicators, and self-evaluators out of all students” (Wiggins 2011). Methods such as these develop social skills, for example, listening skills, conflict resolution, decision-making, problem-solving, negotiation and peer engagement, while also aiding deeper engagement with the subject matter under investigation.

Links to social and emotional learning

There are strong links between social skills and the affective skills associated with self-management. The Collaborative for Academic, Social, and Emotional Learning (CASEL), for example, has identified five core social and emotional competencies (see figure 2).



This approach emphasizes elements such as the ability to take the perspective of others and the ability to form effective relationships. However, it also emphasizes the importance of students developing the ability to regulate their own emotions and behaviours. This links strongly to the affective skills associated with self-management skills.

Related resources

[Video: Social skills: collaboration \(Ho Chi Minh City\)](#)

Name of school: British International School of Ho Chi Minh City, Vietnam

DP subject: Language A literature lesson

[Case study: Pamoja Education developing collaboration skills](#)

[Holistic education: An interpretation for teachers in the IB programmes](#)

Self-management skills

This skill category breaks down into two separate areas.

1. Organization skills—managing time and tasks effectively, goal-setting, etc.
2. Affective skills—managing state of mind, self-motivation, resilience, mindfulness, etc.

Organization skills

One of the most crucial skill sets needed for success in the DP are organizational skills, and within that the particular skill of time management. Students at both secondary and tertiary levels are very aware of their own deficiencies in this area, but often do not have effective strategies to overcome them (Weissberg et al. 1982).

Good time management is a feature of self-managed or self-directed learning (McCombs 1986): it can alleviate stress (Lay and Schouwenburg 1993), increase academic performance (Campbell and Svenson 1992) and contributes significantly to successful “strategic” study (Kirschenbaum and Perri 1982). Time management is not something we can assume that students will do naturally—as with all ATL skills, it is a specific skill that must be taught and also modelled. If we expect our students to be well organized and punctual, to work methodically throughout the year and to meet all deadlines without last minute panic, then we must model strategies to help them achieve this goal. DP teachers can help students organize their time by coordinating their deadlines for students so that assessments are well spread throughout the school year. If teachers also help their students

learn how to break down assignments into achievable steps and to timeline each step, plan out revision and study plans for tests and examinations, and build study timetables, then much poor time management will be alleviated.

Teaching idea

To model good organization and avoid unnecessary student stress, coordinate assessment dates for assignments, tests and internal assessments with colleagues from all other subjects in the DP.

One reason that has been suggested for poor time management among students is perceived *control of time* (Macan, Shahani, Dipboye and Phillips 1990). Some students feel that control over their time is something that is out of their hands, and consequently feel more stressed, procrastinate more and produce poorer quality work. In these situations, it is not the lack of time that is the key factor—it is the perception of control. Performance improvement in this area, therefore, comes partly out of time-management strategies themselves and partly out of attitude and perception, both of which can be influenced by affective skill development.

Affective skills

The development of affective skills is a key part of the development of self-management skills. This can enable students to gain some control over their mood, their motivation and their ability to deal effectively with setbacks and difficulties. There is also an important link between this area of ATL and the crucial area of student health and well-being, which historically has tended to be “mostly separated from other aspects of school life” (Konu and Rimpelä 2002).

Affective self-management skills are teachable and they can make a huge difference to a child’s motivation, resilience and, indeed, academic success; for example, relaxation training can help reduce examination anxiety and increase grades (Hembree 1988). For DP students, three important affective skills that are needed to handle the challenges of this level of study are resilience, self-motivation and mindfulness.

Resilience

Resilience appears to be the affective concept that is most inclusive of almost all other desirable affective elements of the successful DP student. The resilient learner is mindful, persevering, emotionally stable and self-motivated. Through focusing on developing resilience with respect to learning, teachers may find that many other important affective skills are practised and developed as well. The optimal conditions for learning do not seem to be created by goals that are too easy or too difficult, but by goals that are challenging but achievable (Csikszentmihalyi, Rathunde and Whalen 1993). For the resilient learner, any challenge entails the possibility of failure and frustration, but it is this possibility that makes the challenge interesting and intrinsically motivating (Alfi, Assor and Katz 2004).

An important connection to highlight between ATL skills and the learner profile is the connection between resilience and the learner profile attribute of being “risk-takers”. Resilience is a vitally important part of self-management, and it includes learning from mistakes. DP students are often under a great deal of pressure to succeed, particularly given the high stakes nature of DP assessment, which can make them reluctant to ever risk failure. Kathryn Schulz, author of *Being Wrong: Adventures in the Margin of Error*, is critical of the largely negative view of error, commenting that “Of all the things we are wrong about, this idea of error might well top the list. It is our meta-mistake: We are wrong about what it means to be wrong. Far from being a sign of intellectual inferiority, the capacity to err is crucial to human cognition” (Schulz 2011: 5). Risk-taking is therefore closely linked to the idea of failing well (King 2009), and it is important that teachers create an atmosphere where students do not feel that they have to get things right first time. Regarding learning as a process of gradual improvement through reflection on mistakes can encourage students to ask questions, take risks, be more adventurous in their thinking, and be more creative with their ideas.

Self-motivation

Self-motivation is also at the core of successful learning. As Alfie Kohn points out in *Punished by Rewards* (2000), no one can ever really motivate anyone else; the only true motivation is self-motivation (Brandt 1995). All we can ever hope to do as teachers is to arrange the variables that we have some influence over to help self-motivation to arise and then to facilitate its development. Interestingly, the research into intrinsic motivation shows that classroom environments that actively help develop students’ autonomy and self-direction are also those that increase students’ intrinsic motivation and help improve their efficacy as learners (Deci 1975), whereas classroom environments

that use tangible rewards for performance outcomes tend to undermine intrinsic motivation (Deci, Koestner and Ryan 1999).

Mindfulness

Psychological techniques, such as mindfulness, focus on the practice of mental relaxation, and in an educational context there is evidence that mindfulness training can lead to improvements in the functioning of the brain (Brown, Ryan and Cresswell 2007), with improvements being reported in reading comprehension and working memory capacity (Mrazek et al. 2013), digital memory span (Chambers, Lo and Allen 2008) and visual/spatial processing efficiency (Kozhevnikov et al. 2009). *Mindfulness* means simply becoming more aware of your own perceptions as they happen, and in your thinking as it occurs.

Helping students learn how to “stay in the moment” can help them to overcome distractions, increase attention and improve concentration (Brefczynski-Lewis et al. 2007). Most students—at all levels of schooling—and most parents as well, acknowledge that the ability to deal with distractions, and to focus and concentrate on schoolwork is probably the single biggest issue to overcome in order to improve academic performance. Of all the skills we could possibly help our students with, learning to concentrate has to be one of the most important and may be worthwhile putting some training into developing.

Teaching idea

Explore the possibilities for improving students’ concentration through training in mindfulness or other similar techniques.

Related resources

[Video: Approaches to self management \(Toronto\)](#)

Name of school: Branksome Hall, Toronto, Canada

DP subject: Short documentary

[Case study: Bringing mindfulness into the classroom](#)

[Case study: Developing self-management skills](#)

Research skills

“Most students think of researching as putting key words into a search box which leads them to undervalue the importance of other methods.”

— *(Nichols and Mellinger 2007)*

The development of research skills is given a central place in the DP, as can be seen, for example, through the importance placed on the extended essay. The extended essay provides the opportunity for students to undertake personal research into the study of a topic of their own choice, yet with the support and guidance of a supervisor. The extended essay is a demanding task, which is intended to help students to develop sophisticated research and writing skills. Yet the completion of such a task in a school environment, with the support and guidance of a supervisor, is intended to serve as excellent preparation for university studies, where students are likely to have to undertake similar tasks without the same level of structure, scaffolding and support.

Research skills are also at the heart of inquiry-based pedagogy, which heavily influences all IB programmes. “It is acknowledged that this type of pedagogical approach requires advanced information literacy capabilities in students, and that there is a need to support the development of information literacy in inquiry-based learning curricula” (McKinney 2014). (For more information on this approach, please see the “[Teaching based on inquiry](#)” section later in this document.)

There is also an important connection between effective research skills and academic honesty. While the IB does see some cases of students engaging in deliberate academic malpractice, it sees a far greater number of examples of students unintentionally committing malpractice because they, for example, lack research skills such as citation and referencing skills. All DP students are expected to acknowledge fully and in detail the work, thoughts or ideas of another person if they are incorporated in work submitted for assessment. The IB does not prescribe which style(s) of referencing should be used by students; this is left to the discretion of the school. However, it is expected that students will use an appropriate style, and use this consistently. More information on academic honesty can be found in the document *Academic honesty*, available on the online curriculum centre (OCC).

Teaching idea

Promote research skills and critical-thinking skills by explicitly asking students to discuss and reflect on the value and limitations of the resources they chose to use when researching an assignment.

Research skills in the 21st century

Good-quality research skills have always been at the heart of academic endeavour, but the mechanisms used and media of information have changed enormously in the last 30 years. In 1981, Marland broke research skills down into nine sequential stages: formulate and analyse needs; identify and appraise likely sources; locate individual resources; examine, select and reject sources; interrogate sources; record and store information; interpret, analyse, synthesize and evaluate information collected; present and communicate resulting work; and evaluate what has been achieved. All of these skills are just as valid today as they were then; however, the presence of the electronic library and the internet makes some of those skills much broader in application than they once were (Barry 1997).

Research today often requires much more comparing, contrasting and validating available information, and winnowing down the volume of data into a manageable quantity. However, fundamental research skills, such as formulating focused and precise research questions, are as crucial today as they always have been.

Teaching idea

When setting assignments that involve the student giving an oral presentation, use the opportunity to discuss the importance of academic honesty and clear referencing of source materials in all tasks, including oral presentations. Many students forget that it is just as crucial to acknowledge their sources in an oral presentation as it is in a piece of written work. There are a number of ways in which students can effectively do this, such as verbal or written acknowledgments throughout the presentation, or by including a bibliography on the last slide of a PowerPoint®.

For most students today, the independent, self-regulated research involved in inquiry-based learning involves internet-based research. The internet is fast becoming the most important information source in contemporary society, and internet skills can now be considered as vital personal assets. Although they are often seen to be the most prolific of internet users, teenagers are not all equally proficient in the internet skills they need for effective self-directed research. Teenagers often have less well developed search skills than adults, and only rudimentary techniques and strategies for simple searching, hypertext and hypermedia navigation. Four key internet research skills most students need training in are browsing, being aware, searching and monitoring (Bates 2002):

Browsing (or surfing)—this is the skill that most students already have in abundance, characterized by a general initial direction of inquiry followed by a willingness to be distracted in almost any other direction at all. The problem with browsing is that it often takes place when more directed research needs to be carried out.	Being aware—this is more of a critical-literacy skill, and essentially means being aware of all the unsolicited information in our environment, scanning it for relevance but not paying specific or direct attention to it. As a part of DP language and literature courses, students are encouraged to develop this skill.
Searching—using Boolean operators and search limiters to refine searches through search engines of the general type (Google, Yahoo) and the more specific or scholarly type (university library, commercial database, Google Scholar).	Monitoring—employing RSS readers to collect together all internet content (feeds) relevant to school subject lines of inquiry, scanning through all collected feeds on a regular basis looking for topics of value, finding the relevant information and downloading, sharing, posting or filing the important data.

Research and information literacy

Training in information-literacy and media-literacy skills is vital for every student who is engaged in any form of inquiry learning. Feeling overwhelmed by information is a common concern of students, but by developing their research skills students can learn to narrow down the scope of their researching and cope with the volume of information they produce. This is particularly the case with internet-based research, as “In the Internet world, the ease of finding *something* obscures the difficulty of finding the *right thing*” (Gustavson and Nall 2011: 291). In many cases, the people in schools best equipped to help students develop their research skills are librarians. Various strategies may be employed, ranging from librarians team-teaching with subject teachers, offering short courses/seminars, or workshops on specific technological applications and tools. More advice on collaborating effectively with colleagues such as librarians can be found in the “[Enlisting allies: Involving the wider school community](#)” section of this document.

Teaching idea

Find some good online magazines, newspapers, blogs and discussion groups for your subject and encourage all students to set up RSS feeds to gather all the current subject-related ideas. Don't give the students too many, just pick two or three feeds to begin with. Students can then build up their own feeds and share and recommend them between themselves as the year progresses.

Although there is a close link between technological literacy and information literacy, it is important to distinguish between the two. *Information literacy* is a wider term, often used to refer to the complex set of abilities “which enable individuals to engage critically with and make sense of the world and its knowledge, to participate effectively in learning, and to make use of and contribute to the information landscape” (Hepworth and Walton 2009: 10).

It is also important to remember that research skills do not exist in isolation. They are often intimately linked to other skills, such as communication skills, and these can also be developed alongside any research skill practice.

Related resources

[Video: Research skills \(The Hague\)](#)

Name of school: American School of The Hague, Netherlands

DP subject: Theatre lesson

[Case study: Improving Thinking and Research Skills in Visual Arts](#)

[Academic integrity policy](#)

